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EXP6:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int modinv(int a, int m) {
    int x;
    a = ((a % m) + m) % m;
    for (x = 1; x < m; x++)
        if ((a * x) % m == 1) return x;
    return -1;
}

int main() {
    char ct[500], pt[500];
    int freq[26] = {0}, i, a, b, ainv;
    int max1 = 0, max2 = 1;
    int e = 'E' - 'A', t = 'T' - 'A';

    printf("Enter ciphertext: ");
    scanf("%s", ct);
    for (i = 0; ct[i]; i++) freq[toupper(ct[i]) - 'A']++;

    /* find most and second most frequent letters */
    for (i = 0; i < 26; i++)
        if (freq[i] > freq[max1]) max1 = i;
    for (i = 0; i < 26; i++)
        if (i != max1 && (max2 == max1 || freq[i] > freq[max2])) max2 = i;
```

```
    printf("Most frequent letter       : %c\n", max1 + 'A');
    printf("Second most frequent letter: %c\n", max2 + 'A');

    /* assume E -> most frequent, T -> second most frequent
       a*4  + b = max1 (mod 26)
       a*19 + b = max2 (mod 26)  => a = (max2-max1)*inv(15) mod 26 */
    ainv = modinv(t - e, 26);
    if (ainv < 0) { printf("Cannot solve, try another letter pair.\n"); return 0; }
    a = (((max2 - max1) % 26 + 26) % 26 * ainv) % 26;
    b = ((max1 - a * e) % 26 + 26) % 26;
    printf("Solved key: a = %d, b = %d\n", a, b);

    ainv = modinv(a, 26);
    for (i = 0; ct[i]; i++) {
        int c = toupper(ct[i]) - 'A';
        pt[i] = (ainv * ((c - b + 26) % 26)) % 26 + 'a';
    }
    pt[i] = '\0';
    printf("Decrypted text: %s\n", pt);
    return 0;
}
```

OUTPUT:

```
Enter ciphertext: UKBBGPZNCPUNFCFEBABOJBCVOJIUBYZBRRPHBOBLXNOBRIPUNBCVBPCYVPOBPCYUKBUBPVKBORBCU
UKBBRRBCUNPWCFUBRUFUKBRUXYBCURUKNRBABCNCH
Most frequent letter       : B
Second most frequent letter: U
Solved key: a = 3, b = 15
Decrypted text: theexaminationofeveryencryptedmessagerequirespatienceandcareandtheteachersentth
eessentialnotestothestudentsthisevening
```